

DAVID MENDOZA

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EDUCATION

University of Houston, Cullen College of Engineering — Houston, Texas

Bachelor of Computer Engineering Technology | Dean's List

December 2026

TECHNICAL SKILLS

- **Programming:** Python, Verilog, C, C++.
- **Hardware, Tools & Frameworks:** PyTorch, CUDA, LeRobot, I2C, FPGA, MATLAB, Quartus, LTspice.
- **Course Work:** Digital Systems, Microprocessor Architecture, AI in Robotics, Computer Networks, Unix OS.
- **Languages:** English (Native/Fluent), Spanish (Native/Fluent), French (Conversational / B1).

PROJECTS

AI Robotic Arm

- Developed an AI-driven robotic arm using an NVIDIA Jetson Orin Nano, leveraging LeRobot and CUDA integration to accelerate machine learning model training for automated robotics tasks.
- Interfaced sensor arrays with L298N motor controllers, implementing PID control algorithms to ensure precise, real-time physical adjustments and smooth joint movements.

Reaction Timer

- Engineered a digital reaction timer on a Terasic Altera Cyclone IV FPGA, writing modular VHDL/Verilog code for a finite state machine (FSM) to measure and display human response latency with millisecond precision.

Tiny-VGG

- Built and optimized a custom CNN based on the VGG architecture using PyTorch and CUDA. The model was trained on a sample Kaggle image classification dataset (cats and dogs) using an NVIDIA GPU, achieving 80% test accuracy.

EXPERIENCE

TOC/CIR Assistant | FIFA World Cup 2026

June 2026 – July 2026

FIFA - hbs, Houston, Texas

- Facilitated the integration of Media Partners' (MPs) broadcasting equipment with venue infrastructure, routing complex fiber optic looms through the Center Interconnection Room (CIR).
- Monitored live feed stability and signal distribution across global networks, acting as a critical technical checkpoint between the Technical Operations Centre (TOC) and Technical Interconnection Room (TIR).
- Ensured uninterrupted broadcast operations by verifying that all media partners were successfully sending and receiving high-quality feeds.
- Collaborated with NEP and ARCTEK engineering teams to validate Outside Broadcast (OB) truck signal processing and calibrate satellite uplink transmissions.

Paint Associate

July 2020 - Present

The Home Depot, Houston, Texas

- Balanced 20+ hours weekly alongside full-time engineering coursework, managing high-volume inventory logistics and translating technical chemical/coating specifications for B2B contractors and retail clients.